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Mayank Raj

Ph.D. Research Summary

My PhD research lies at the intersection of machine learning, applied mathematics, and material science. I develop data-driven and physics-informed machine-learning methods for discovering models of complex materials using computational and experimental data. The particular focus of my research is on learning governing laws, constitutive relations, and latent variables while incorporating physical structure and mathematical constraints. Find my Ph.D candidacy slides here and my projects portfolio here.

Education

California Institute of Technology — Pasadena, CA, USA
*Ph.D. in **Applied and Computational Mathematics*** — 2022 – 2027
- ❑ Advisor: Prof. Andrew M. Stuart
- ❑ GPA: 4.1

Indian Institute of Technology Madras — Chennai, India
*M.Tech. in **Data Science*** — 2020 – 2021
*B.Tech. in **Mechanical Engineering*** — 2016 – 2020
- ❑ Minor in **Applied Mathematics**
- ❑ GPA: 9.28/10

Employment History

Wells Fargo — Bangalore, India
Data Scientist — 2021 – 2022
- ❑ Developed an **automated OCR and regular-expression pipeline** to detect, extract, and structure data from unstructured PDF documents.
- ❑ Deployed the pipeline on the firm's **parallel computing** cluster, automating the end-to-end workflow from PDF ingestion and data extraction to structured-data generation.

Publications

A full list of publications on Google Scholar.

[5] **Mayank Raj**, Lianghao Cao, Andrew Stuart, and Kaushik Bhattacharya. "A neural-network framework to learn history-dependent constitutive laws and identifiability of internal variables". *Journal of the Mechanics and Physics of Solids*. Paper Link, Project Website.
- ❑ Built a causal, energy-based neural-network framework for learning history-dependent constitutive laws that is thermodynamically consistent and stable by construction.
- ❑ Showed empirically, and proved under controllability assumptions, that the learned internal variables are identifiable up to a linear transform.
- ❑ Learned the averaged response of a polycrystalline magnesium cell with 2% relative error, replacing costly multiscale (FE^2) simulations.

[4] Harkirat Singh, **Mayank Raj**, and Kaushik Bhattacharya. "Effective behavior of heterogeneous media governed by strain gradient elasticity". *Journal of the Mechanics and Physics of Solids*. Paper Link.
- ❑ Showed through numerical experiments that strain gradient elasticity is not preserved under averaging of heterogeneous media.
- ❑ Described the effective behavior with fractional strain gradient elasticity, giving scaling laws between classical and strain-gradient limits.
- ❑ Built a data-driven model of the effective behavior across length scales using a Fourier neural operator.

[3] **Mayank Raj**, Pramod Kumbhar, and Ratna Kumar Annabattula. "Physics-informed neural networks for solving thermo-mechanics problems of functionally graded material". *arXiv preprint*. Paper Link, Project Website.

- ❑ First application of physics-informed neural networks (PINNs) to coupled thermo-mechanics problems in functionally graded materials.
- ❑ Studied how training data, network architecture and loss-function design affect PINN accuracy.
- ❑ Reached $R^2 > 99\%$ for displacement and temperature against analytical and finite element solutions.

[2] **Mayank Raj**, Sanket Thakre, Ratna Kumar Annabattula, and Anand K. Kanjarla. "Estimation of Local Strain Fields in Two-Phase Elastic Composite Materials Using UNet-Based Deep Learning". *Integrating Materials and Manufacturing Innovation*. Paper Link, Project Website.

- ❑ Trained a UNet deep-learning model to predict full local strain fields in two-phase composites under uniaxial tension.
- ❑ Generated synthetic microstructure datasets to train and test the model.
- ❑ Analyzed how dataset design affects the robustness of deep-learning models for strain localization.

[1] **Mayank Raj**, Sandeep P. Patil, and Bernd Markert. "Mechanical Properties of Nacre-Like Composites: A Bottom-Up Approach". *Journal of Composites Science*. Paper Link, Project Website.

- ❑ Computed the shear modulus of nacre's organic matrix (1.25–1.45 GPa) with atomistic molecular dynamics simulations.
- ❑ Ran 3D finite element simulations of nacre-like composites while varying the mineral and organic composition.
- ❑ Found that 10–20% organic content by volume gives high toughness, guiding the design of synthetic nacre-like materials.

Internships

Tata Research Design and Development Center — Pune, India
Research Intern in Machine Learning, Project Website — Winter, 2020

- ❑ Developed a **machine-learning-driven framework for multi-objective optimization** of product designs, integrating CFD simulations with surrogate modeling.
- ❑ Built a **machine learning surrogate** model to rapidly predict separation factor and pressure drop, enabling efficient exploration of the particle-classifier design space.
- ❑ Performed **multi-objective optimization** to identify Pareto-optimal designs balancing separation performance against pressure drop.

RWTH Aachen University — Aachen, Germany
Research Intern in Computational Mechanics, Project Website — Summer, 2019

- ❑ Modeled the 3D brick-and-mortar microstructure of nacre using HyperMesh, representing the composite architecture of brittle calcium carbonate and ductile organic matrix.
- ❑ Conducted LS-DYNA simulations to characterize the mechanical properties and investigate the effect of material composition on the toughness of nacre-based composites.
- ❑ Demonstrated that increasing the organic-matrix fraction enhances composite toughness but results in a corresponding reduction in overall strength.

Awards & Achievements

- ❑ The Bell Family Fellowship — 2022
- ❑ Mitacs Globalink Scholarship — 2020
- ❑ UGC-DAAD Research Scholarship — 2019
- ❑ Top 0.15% in JEE-Advanced — 2016

Teaching Assistant

Linear Analysis with Applications; Signal-Processing Systems and Transforms; Introductory Methods of Applied Mathematics for the Physical Sciences; Introduction to Computational Science & Engineering

Relevant Coursework	Probability, Statistics & Stochastic Processes; Stochastic Differential Equations; Linear Algebra; Optimization; Deep Learning	
Conferences & Workshops	❑ Frontiers in Applied Analysis, Carnegie Mellon University	2025
	❑ Machine Learning Theory Summer School, Princeton University	2025
Tools	PyTorch, Numpy, Pandas, Scipy, Scikit Learn	

References

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